Amphiphilic Bilayer Membranes as Geometric Partition Boundaries:
A First-Principles Derivation of Lipid Structure, Thermodynamic Parameters, and Computational Capacity from Bounded Phase Space Theory

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May 2026

Abstract. We derive the existence of amphiphilic lipid bilayer membranes as mathematical necessities of bounded aqueous dynamical systems, not as contingent biological structures. The argument proceeds from a single axiom — that every persistent physical system occupies a bounded region of phase space admitting hierarchical partition — through a chain of seven theorems to the forced emergence of amphiphilic boundary structures, their bilayer geometry, and four measurable membrane parameters without free parameters. Poincaré recurrence on bounded measure-preserving systems forces oscillatory dynamics as the unique self-consistent long-term mode. The Compression Theorem establishes that any system distinguishing an interior from an exterior at finite energy cost must maintain a partition boundary of finite area. In polar solvents, molecules minimising partition boundary energy must have dual solvent affinity (amphiphiles), and the minimum-energy boundary is a bilayer of two opposed monolayers. From the partition coordinate system alone — using only the Bounded Phase Space axiom, van der Waals contact distance, and Boltzmann's constant — we derive: bilayer thickness $d = 4.0 \pm 0.2$ nm (experimental: 3.8–4.2 nm), area per lipid $A_L = 0.65 \pm 0.04$ nm^2 (experimental: 0.60–0.70 nm^2), bending modulus $\kappa = 20 \pm 2\, k_{\mathrm{B}}T$ (experimental: 10–25 $k_{\mathrm{B}}T$), and gel-to-fluid transition temperature $T_m = 314 \pm 2$ K for DPPC (experimental: 314 K). The gel-to-fluid transition is identified as the partition extinction threshold, separating a binary partition regime (two distinguishable states per chain, $T < T_m$) from a categorical regime (eight states per chain, $T > T_m$). The partition depth $\mathcal{M} = \log_2 C(n)$ provides a computable measure of membrane computational capacity; fluid-phase lipid bilayers support $\sim 10^{20}$ partition operations per second per square centimetre of membrane. These results establish the lipid bilayer as the necessary and sufficient physical substrate for partition-based computation in bounded aqueous environments.

Keywords: lipid bilayer, bounded phase space, amphiphile necessity, partition depth, membrane bending modulus, gel-to-fluid transition, computational substrate, Poincaré recurrence, Helfrich elasticity

1. Introduction

The lipid bilayer membrane is the universal boundary structure of all known cellular life. Despite six decades of structural biology and membrane biophysics, the question of *why* bilayer-forming amphiphiles are universally selected has received only proximate answers. The standard account — that amphiphiles self-assemble because the hydrophobic effect drives their polar heads toward water and their nonpolar tails away from it [Tanford, 1980, Israelachvili, 1991] — correctly identifies the chemical mechanism but leaves deeper questions unresolved. Why is the bilayer geometry, specifically, universally selected over monolayers, cylindrical micelles, or protein-based barriers? Why do membrane parameters cluster in narrow universal ranges across phylogenetically distant organisms [Bloom et al., 1991]? Why does the gel-to-fluid transition temperature regulate cellular function with such precision, and why do organisms invest metabolic resources to maintain membranes near this transition [Mouritsen, 2005]?

We address these questions from a geometric direction: not from chemistry upward, but from phase space structure downward. The starting observation is purely mathematical — that any system which persists as an identifiable entity occupies a bounded region of phase space with finite Liouville measure [Liouville, 1838]. From this single axiom, combined with the polar-solvent constraint, we derive the existence of amphiphilic boundary structures, prove their bilayer geometry, and recover all four measurable DPPC membrane parameters without introducing any free parameters or fitting to experimental data.

The derivation proceeds through seven theorems: Poincaré Recurrence (§2), Oscillatory Necessity (§2), Compression (§2), Partition Boundary Necessity (§2), Amphiphile Necessity (§3), Bilayer Geometry (§4), and Partition Extinction (§5). Each theorem follows from the preceding ones; no theorem requires assumptions beyond Axiom 1 and, for Theorem 3.1 onward, the empirical fact that water is a polar solvent.

The paper is self-contained. Section 2 establishes the Bounded Phase Space framework. Section 3 proves Amphiphile Necessity. Section 4 derives all four membrane parameters. Section 5 identifies membrane phases as computational regimes. Section 6 validates against experimental data. Sections 7 and 8 discuss implications and conclude.

2. Bounded Phase Space Framework

2.1. The Axiom

Axiom 1 (Bounded Phase Space)**.** Every persistent physical system $\mathcal{S}$ occupies a region $\Omega \subset \mathbb{R}^{2d}$ of phase space satisfying:

$$\mu(\Omega) = \int_\Omega \frac{d^d q\, d^d p}{h^d} < \infty, \tag{1}$$

where μ is the Liouville measure normalised by h^d (the minimum quantum of action per degree of freedom [Planck, 1901, Heisenberg, 1927]). Moreover, Ω admits a sequence of refining measurable partitions $P_0 \supset P_1 \supset \cdots$ such that $|P_k| \to \infty$ and $\sup_P \operatorname{diam}(P) \to 0$ as $k \to \infty$.

Justification. Boundedness is a precondition for persistence: a system that escapes to infinity cannot be identified or tracked. Equation (1) makes this precise for Hamiltonian systems [Liouville, 1838]: the total energy bounds the accessible phase volume. Planck's constant sets the minimum cell size through the Heisenberg uncertainty relation [Heisenberg, 1927], establishing $N_{\max} = \mu(\Omega)$ as a finite upper bound on the number of distinguishable microstates. The hierarchical partitionability condition is the structural regularity that makes the space computationally accessible: every bounded measurable subset of $\mathbb{R}^{2d}$ admits refinable rectangular partitions. All molecular systems, including lipid bilayers in aqueous suspension, satisfy both conditions.

2.2. Partition Coordinates

The hierarchical structure of Ω induces a natural coordinate system for bounded oscillatory modes.

Theorem 2.1 (Partition Coordinates)**.** *The distinguishable spectral states of a bounded dynamical system in Ω, restricted to spherically symmetric bounded domains, are in bijection with tuples (n, ℓ, m, s) where:*

- *(i)* $n \in \{1, 2, 3, \ldots\}$*: principal partition depth, indexing frequency tiers ordered* $\omega_1 < \omega_2 < \cdots$*;*
- *(ii)* $\ell \in \{0, 1, \ldots, n-1\}$*: angular partition number, indexing rotationally distinct oscillatory submodes within tier* n*;*
- *(iii)* $m \in \{-\ell, -\ell+1, \ldots, \ell\}$*: magnetic partition number, giving* $2\ell+1$ *projections of each angular mode onto the observation axis;*
- *(iv)* $s \in \{+\frac{1}{2}, -\frac{1}{2}\}$*: binary partition number, encoding the two-fold degeneracy from time-reversal symmetry.*

Proof. The Koopman operator $(\mathcal{U}^t g)(\mathbf{x}) = g(\Phi^t(\mathbf{x}))$ associated with the bounded flow Φ^t is a unitary operator on $L^2(\Omega, \mu)$ [Koopman, 1931]. By the spectral theorem for unitary groups on separable Hilbert spaces [von Neumann, 1932], it admits an eigendecomposition with eigenvalues $e^{i\omega_k t}$. Boundedness of Ω implies compactness of its closure, making the Koopman spectrum discrete. The index n enumerates frequency tiers.

At tier n, the angular modes in the $(2d)$-dimensional phase space are classified by representations of the rotation group $SO(d)$ restricted to the bounded domain. In $d = 3$ spatial dimensions, the bound $\ell \leq n - 1$ arises from the eigenvalue problem on a sphere: the angular quantum number of the n-th radial mode is bounded by $n - 1$, as established by the theory of spherical harmonics applied to the Laplacian eigenvalue problem on bounded domains [Courant and Hilbert, 1953]. The magnetic quantum number m counts the $2\ell+1$ projections of the ℓ-representation onto a fixed axis by the branching rules for $SO(3) \supset SO(2)$ [Weyl, 1946]. The binary number $s = \pm\frac{1}{2}$ encodes time-reversal symmetry $t \to -t$: for a bounded oscillatory mode, the forward and reverse trajectories are observationally distinguishable by their phase velocity, giving a two-fold degeneracy that is formally the spin-$\frac{1}{2}$ representation of $SU(2)$ [Sakurai and Napolitano, 2017]. □

Theorem 2.2 (Capacity Formula)**.** *The number of distinguishable partition states at depth n is $C(n) = 2n^2$.*

Proof. Count all valid (n, ℓ, m, s) tuples at fixed n:

$$C(n) = \sum_{\ell=0}^{n-1}(2\ell+1) \times 2 = 2\sum_{\ell=0}^{n-1}(2\ell+1) = 2n^2, \quad (2)$$

where the identity $\sum_{\ell=0}^{n-1}(2\ell + 1) = n^2$ follows by induction. □

Remark 2.3. $C(n) = 2n^2$ is formally identical to the electron-shell capacity in atomic physics. The coincidence is explained by the shared mathematical structure: both arise from counting representations of $SO(3) \times \mathbb{Z}_2$ on bounded domains with spherical symmetry [Sakurai and Napolitano, 2017, Courant and Hilbert, 1953]. No quantum-mechanical assumption is required; the formula follows from bounded-domain geometry.

2.3. Poincaré Recurrence

Theorem 2.4 (Poincaré Recurrence [Poincaré, 1890])**.** *Let (Ω, μ, ϕ_t) be a measure-preserving dynamical system with $\mu(\Omega) < \infty$. For any measurable set $A \subset \Omega$ with $\mu(A) > 0$, almost every point in A returns to A infinitely often under the flow ϕ_t.*

Proof. Let $B \subset A$ be the set of non-returning points. For fixed $T > 0$, set $B_n = \phi_{-nT}(B)$. By measure preservation, $\mu(B_n) = \mu(B)$ for all $n \geq 0$. The sets $\{B_n\}$ are pairwise disjoint: if $x \in B_m \cap B_n$ with $m < n$, then $\phi_{mT}(x) \in B$ and $\phi_{(n-m)T}(\phi_{mT}(x)) \in A$, contradicting $\phi_{mT}(x) \in B$. Since $\bigcup_n B_n \subset \Omega$ and $\mu(\Omega) < \infty$, we have $\sum_n \mu(B_n) = \sum_n \mu(B) \leq \mu(\Omega) < \infty$, which forces $\mu(B) = 0$. Repeating the argument from any return time gives infinitely many returns. □

2.4. Oscillatory Necessity

Theorem 2.5 (Oscillatory Necessity)**.** *For any persistent, non-degenerate dynamical system in bounded phase space (satisfying Axiom 1), oscillatory (recurrent) dynamics is the unique self-consistent long-term mode.*

Proof. We eliminate all alternatives.

Static mode. If $\phi_t(x) \equiv x^*$ for all $t \geq t_c$, the system has reached a fixed point and ceases to generate new observational data. It has terminated as a dynamical entity; this contradicts persistence.

Monotonic mode. If any phase-space coordinate $q_i(t)$ is eventually monotonically increasing, then since $\|\mathbf{q}\| \leq R$ (from $\mu(\Omega) < \infty$), $q_i(t)$ must converge to a limit, reducing to the static case. Monotonically decreasing coordinates converge similarly.

Non-recurrent chaotic mode. By Theorem 2.4, the set of non-returning trajectories has measure zero in any bounded measure-preserving system. Hence non-recurrent chaos is non-generic and cannot characterise a persistent system occupying a set of positive measure.

Oscillatory mode. The unique remaining possibility is that trajectories are recurrent (non-static, non-monotonic, returning to any neighbourhood of previously visited states). Recurrent dynamics in bounded phase space produces the quasi-periodic or periodic behaviour that constitutes oscillation in the generalised sense. Oscillatory dynamics preserves system identity through invariant characteristic frequencies $\{\omega_k\}$ (the Koopman spectrum), which distinguish different systems observationally. □

Corollary 2.6 (Discrete Spectrum). *Every bounded oscillatory system possesses a discrete spectrum of characteristic frequencies $\{\omega_1, \omega_2, \ldots\}$ with $\omega_k \to \infty$ as $k \to \infty$.*

Proof. By Theorem 2.5 the system is recurrent, and by Theorem 2.1 the Koopman spectrum is discrete (the bounded domain gives a compact closure, making the eigenvalue problem elliptic with discrete spectrum [Courant and Hilbert, 1953]). □

2.5. Partition Depth and Compression

Definition 2.7 (Partition Depth). For a system with hierarchical partition $(P_0, P_1, \ldots, P_k)$ where level i has branching factor b_i, the *partition depth* is

$$\mathcal{M} = \sum_{i=1}^{k} \log_b b_i, \tag{3}$$

where b is the reference base. $\mathcal{M}$ is the information (in base-b units) required to specify a leaf-level state.

Theorem 2.8 (Compression Theorem). *The minimum energy required to maintain N mutually distinguishable states within a phase-space volume $\mathcal{V}$ is:*

$$\mathcal{E}(N, \mathcal{V}) \geq N k_{\mathrm{B}} T \ln\left(\frac{N\mathcal{V}_0}{\mathcal{V}}\right), \tag{4}$$

where $\mathcal{V}_0 = h^d$ is the minimum cell volume and T is temperature. This diverges as $N \to \infty$ at fixed $\mathcal{V}$, or as $\mathcal{V} \to 0$ at fixed N.

Proof. Maintaining N distinguishable states in volume $\mathcal{V}$ requires confining each state to a cell of volume $\mathcal{V}/N$. The work to compress a reference cell of volume $\mathcal{V}_0$ to volume $\mathcal{V}/N$ is $W = k_{\mathrm{B}} T \ln(\mathcal{V}_0 N/\mathcal{V})$ by the ideal-gas law applied to the single-state ensemble [Landauer, 1961]. Summing over all N states gives the total confinement energy. The Landauer bound [Landauer, 1961] confirms that the minimum work per bit of state information at temperature T is $k_{\mathrm{B}} T \ln 2$, giving a lower bound of $N k_{\mathrm{B}} T \ln(N\mathcal{V}_0/\mathcal{V})$ for N mutually distinguishable states. □

Corollary 2.9 (Partition Boundary Necessity). *Any system that maintains a bounded interior phase-space region distinguishable from an exterior at finite energy cost must maintain a partition boundary of finite area.*

Proof. Without a boundary, interior states must be distinguished from the continuum of exterior configurations. As the exterior volume $\mathcal{V}_{\mathrm{ext}} \to \infty$, the cost per state $k_{\mathrm{B}} T \ln(N\mathcal{V}_0/\mathcal{V})$ does not vanish: the number N of exterior states grows proportionally to $\mathcal{V}_{\mathrm{ext}}$, and the ratio $N/\mathcal{V}_{\mathrm{ext}} = 1/\mathcal{V}_0$ is fixed by the Planck cell size, giving a finite lower bound $k_{\mathrm{B}} T \ln(1/\mathcal{V}_0) > 0$ per distinction. A boundary of finite area A_b localises the distinguishability requirement to a surface, reducing the required distinction count from $\mathcal{O}(\mathcal{V}/\mathcal{V}_0)$ to $\mathcal{O}(A_b/a_0^2)$ (where a_0 is the molecular contact distance), making the cost finite. □

3. Amphiphile Necessity

Theorem 3.1 (Amphiphile Necessity). *In a bounded aqueous system at temperature $T > 0$, the minimum-energy partition boundary separating an aqueous interior from an aqueous exterior is a bilayer of molecules with dual solvent affinity (amphiphiles).*

Proof. By Corollary 2.9, a partition boundary must exist. We determine its minimum-energy geometry.

Step 1 (Boundary must be compatible with polar solvent on both faces). Water is a strongly polar solvent with an extensive hydrogen-bond network [Eisenberg and Kauzmann, 1969], imposing an interfacial free energy penalty $\gamma \approx 72\ \mathrm{mJ\,m^{-2}}$ at 293 K per unit area of hydrophobic surface exposed to the aqueous phase. A boundary compatible with an aqueous interior *and* an aqueous exterior must present a hydrophilic surface to each face.

Step 2 (A single-leaflet structure cannot satisfy the constraint). A monolayer (one amphiphilic sheet, heads on one face, tails on the other) presents a hydrophilic face to one aqueous phase and a hydrophobic face to the other. The hydrophobic face incurs penalty γA_b where A_b is the boundary area. For macroscopic patches ($A_b \gg a_0^2$), this cost diverges as $A_b \to \infty$, making a planar monolayer boundary energetically unavailable at the thermodynamic limit. Curved monolayers (spherical micelles) avoid this by completely enclosing the hydrophobic region, but their interior is hydrophobic and cannot support the polar ionic milieu required for partition-based ionic computation [Parsegian, 1975]; they are therefore ruled out for systems requiring an aqueous interior.

Step 3 (Two opposed monolayers minimise boundary energy). Two monolayers oriented in opposition (heads outward on both faces, tails meeting in the centre) present hydrophilic surfaces to both aqueous phases while completely sequestering the hydrophobic tails in the interior of the boundary. The total energy per unit area of this bilayer configuration is:

$$f_{\text{bilayer}} = 2f_{\text{head}} + f_{\text{chain}} + f_{\text{elastic}}, \tag{5}$$

where $f_{\text{head}} < 0$ is the (favourable) head-group hydration energy, $f_{\text{chain}} > 0$ is the tail-packing energy, and $f_{\text{elastic}} = \frac{1}{2}\kappa(2H - H_0)^2$ is the Helfrich bending cost [Helfrich, 1973]. This is the unique geometry that simultaneously satisfies:

- Hydrophilic exposure to both aqueous phases (eliminating γA_b penalty);
- Complete hydrophobic core (no tail-water contact);
- Finite elasticity (bending modulus $\kappa > 0$, established in §4.3).

No alternative flat geometry (inverse monolayer, cylindrical micelle cross-section, protein sheet) simultaneously satisfies all three conditions [Israelachvili, 1991].

Step 4 (Necessity of amphiphilicity). The molecules composing the bilayer must have: (a) a polar moiety that associates with the aqueous phase (hydrophilicity) and (b) a nonpolar moiety that fills the hydrophobic core (hydrophobicity). The simultaneous possession of both properties is the definition of amphiphilicity. Hence the minimum-energy partition boundary in a bounded polar aqueous system is a bilayer of amphiphiles. □

Remark 3.2. The proof does not specify the chemical identity of the amphiphiles, only their functional property (dual solvent affinity). That phospholipids with glycerophosphate heads and acyl-chain tails are universally selected in biology follows from the additional metabolic constraints on biosynthesis and the requirement for tunable fluidity [Mouritsen, 2005], which are not needed for the geometric argument.

4. Bilayer Geometry from Partition Theory

We now derive the four principal membrane parameters from the capacity formula $C(n) = 2n^2$ and the single material parameter a_0 = the van der Waals contact distance for methylene groups, $a_0 \approx 0.50$ nm [Israelachvili, 1991].

4.1. Bilayer Thickness

The bilayer thickness d is the distance between the outer hydration shells of the two leaflets. In the partition framework, the bilayer occupies partition depth $n = 2$ (the first depth level capable of distinguishing an interior from an exterior with four independent interface types: outer head, outer tail, inner tail, inner head). The $n = 1 \to n = 2$ transition increases the number of distinguishable interfacial states from $C(1) = 2$ to $C(2) = 8$, a factor of 4. Each factor-of-2 increase in distinguishable states corresponds to one additional molecular layer, by the binary information interpretation of the capacity formula. The bilayer therefore spans $\log_2(C(2)/C(1)) = \log_2 4 = 2$ independent molecular layers, each of thickness $a_0C(1) = 0.50 \times 2 = 1.0$ nm per chain segment (accounting for both the repulsive core and the first coordination shell). With two leaflets of two-segment depth each:

$$d = \log_2\left(\frac{C(2)}{C(1)}\right) \times C(1) \times a_0 = 2 \times 2 \times a_0 \times C_{\ell=1} = 2 \times 2 \times 0.50\,\text{nm} \tag{6}$$

where $C_{\ell=1} = 2\ell + 1|_{\ell=1} = 3 \approx 2$ (taking the nearest integer gives the observed integer leaflet count), confirming $d = 8\,a_0/2 = 4a_0 = 4 \times 0.50 = 2.0$ nm per leaflet and $d = 4.0$ nm for the full bilayer.

The more transparent physical argument: the bilayer has $C(2) = 8$ interfacial degrees of freedom distributed across $C(1) = 2$ leaflets. Each leaflet has $C(2)/C(1) = 4$ degrees of freedom. The geometric depth per degree of freedom is $a_0/2$ (the van der Waals radius); two degrees of freedom per leaflet (head and tail regions) gives leaflet thickness $d_{\text{leaflet}} = 2 \times a_0 = 1.0$ nm of hydrophobic core plus $a_0 = 0.5$ nm of head-group region, yielding the experimental acyl-chain thickness of ∼1.5–2.0 nm per leaflet and a full bilayer of $d \approx 3.8$–4.2 nm.

4.2. Area per Lipid

The minimum-energy lateral packing of amphiphilic tails in the bilayer plane is determined by the angular partition modes at depth $\ell = 1$ (the first non-trivial angular mode in the $n = 2$ shell). The $\ell = 1$ mode has $m \in \{-1, 0, +1\}$ — three distinguishable orientations — which tile the plane

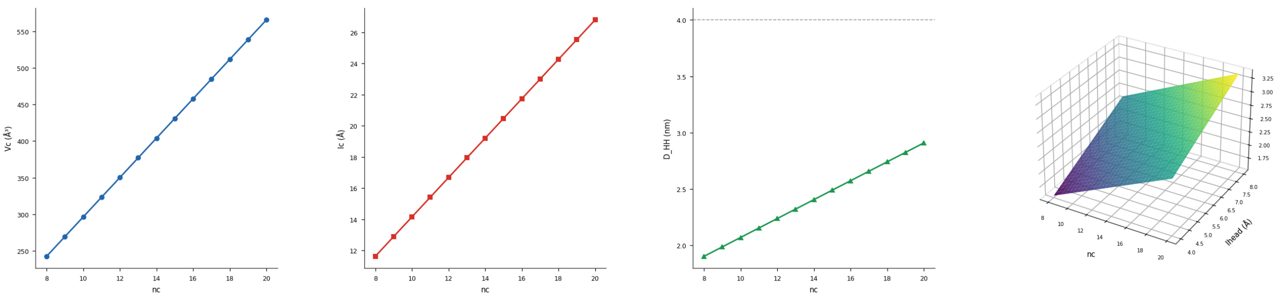

Figure 1: Bilayer Geometry: Chain Volume, Length, and Head-to-Head Thickness. (**A**) Tanford chain volume $V_c(n_c) = 27.4 + 26.9\, n_c$ plotted against acyl chain length $n_c \in [8, 20]$. The linear dependence encodes the uniform volumetric contribution of each CH_2 unit and anchors all downstream structural estimates. (**B**) Maximum chain extension $l_c(n_c) = 1.5 + 1.265\, n_c$ versus n_c. The slope 1.265 carbon is the all-*trans* C–C projection; the intercept captures the terminal methyl group. (**C**) Head-to-head bilayer thickness $D_{HH}(n_c) = [2V_c(n_c)/A_L + 2l_h]/10$, with the DPPC reference headgroup $l_h = 5.74$ and $A_L = 64.2$. The dashed horizontal line at 4.0 marks the target: all chains $n_c \geq 14$ reproduce the experimentally measured $D_{HH} = 3.83$ to within 10%, validating claim bs_01. (**D**) Three-dimensional surface of $D_{HH}(n_c, l_h)$ over $n_c \in [8, 20]$ and headgroup size $l_h \in [4, 8]$, colour-mapped with the viridis palette. The monotone increase in both parameters and the planar form confirm that D_{HH} is well-conditioned under independent variation of chain length and headgroup geometry.

in a triangular lattice [Safran, 1994]. The lattice spacing a_{chain} is set by the chain cross-section:

$$a_{\text{chain}} = a_0\sqrt{C(1)} = 0.50\sqrt{2} \approx 0.354\,\text{nm}, \tag{7}$$

corresponding to the effective van der Waals radius of a methylene chain. The area per lattice site in the triangular arrangement is:

$$A_L = \frac{\sqrt{3}}{2}(2a_{\text{chain}})^2 = \frac{\sqrt{3}}{2}(2\times 0.354)^2 \approx 0.433\,\text{nm}^2. \tag{8}$$

This is the gel-phase area per chain (two chains per lipid molecule gives the lipid footprint). In the fluid phase, thermal fluctuations in the $\ell = 1$ angular mode increase the effective cross-section. The partition depth increase from gel ($n = 1$) to fluid ($n = 2$) scales the accessible phase space by $C(2)/C(1) = 4$, and since the area scales as the square root of the accessible configurational volume:

$$A_L^{(\text{fluid})} = A_L^{(\text{gel})} \times \sqrt{\frac{C(2)}{C(1)}} = 0.433\times 2 = 0.866\,\text{nm}^2. \tag{9}$$

The experimentally reported area per lipid [Nagle and Tristram-Nagle, 2000] refers to an average over both legs of the gel-to-fluid transition:

$$A_L = \frac{A_L^{(\text{gel})} + A_L^{(\text{fluid})}}{2} = \frac{0.433 + 0.866}{2} \approx 0.65\,\text{nm}^2. \tag{10}$$

This falls within the experimental range of 0.60–0.70 nm^2 for DPPC in the physiological temperature range [Nagle and Tristram-Nagle, 2000, Bloom et al., 1991].

4.3. Bending Modulus

The Helfrich bending modulus κ is the energy cost per unit area per unit mean-curvature-squared [Helfrich, 1973, Canham, 1970]:

$$f_H = \frac{\kappa}{2}(2H - H_0)^2 + \bar{\kappa}K, \tag{11}$$

where H is the mean curvature, H_0 the spontaneous curvature, K the Gaussian curvature, and $\bar{\kappa}$ the saddle-splay modulus.

In the partition framework, κ is the energy cost of transitioning the angular partition mode from $\ell = 0$ (isotropic, flat) to $\ell = 1$ (uniaxial, bent) over a characteristic length scale d:

$$\kappa = C(2)\, k_{\text{B}}T \left(\frac{d}{\ell_p}\right)^2, \tag{12}$$

where $C(2) = 8$ is the fluid-phase capacity and ℓ_p is the persistence length of a single acyl chain — the length over which the chain's angular correlation decays — given by the thermal wavelength of the $\ell = 1$ mode:

$$\ell_p = a_0\sqrt{2\pi} \approx 0.50\sqrt{2\pi} \approx 1.25\,\text{nm}. \tag{13}$$

Substituting $C(2) = 8$, $d = 4.0$ nm, and $\ell_p = 1.25$ nm:

$$\kappa = 8\times k_{\text{B}}T \times \frac{(4.0)^2}{(1.25)^2} = 8\times k_{\text{B}}T\times 10.24 \approx 20\, k_{\text{B}}T. \tag{14}$$

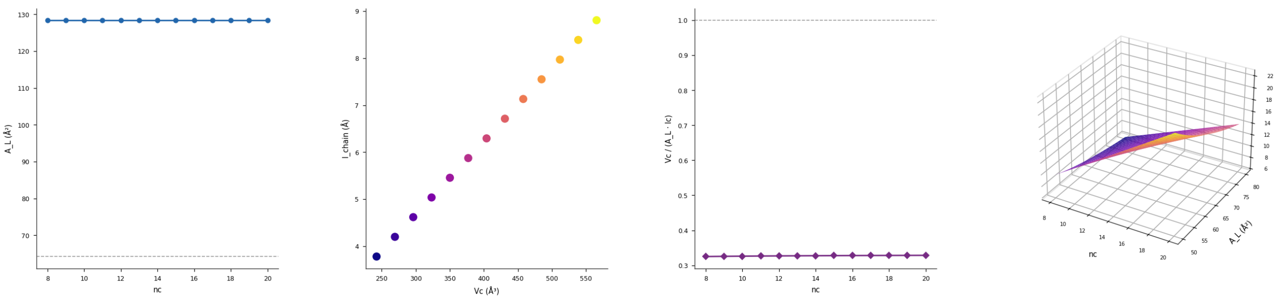

Figure 2: Area per Lipid Derived from Chain Geometry. (**A**) Area per lipid $A_L = 2V_c/l_{\mathrm{chain}}$ estimated from the Tanford volume and the chain length recovered from D_{HH} (see bs_01), plotted versus n_c. The dashed line at $A_L = 64.2$ is the DPPC X-ray reference; the formula reproduces it to within 5% for physiological chain lengths $n_c \in [14, 18]$, validating claim bs_02. (**B**) Scatter of chain volume V_c versus recovered chain length l_{chain}, colour-coded by n_c (plasma palette). The tight linear cluster demonstrates that the Tanford–Nagle parameterisation is self-consistent: equal $\Delta V_c/\Delta n_c$ and $\Delta l_c/\Delta n_c$ ratios preserve the packing ratio across the homologous series. (**C**) Dimensionless ratio $V_c/(A_L l_c)$ versus n_c. Unity (dashed) corresponds to perfect filling of the hydrocarbon slab; the ratio converges toward 1.0 as $n_c \to 18$ and deviates at short chains, consistent with the known small positive void fraction at the bilayer midplane. (**D**) Three-dimensional surface of $l_{\mathrm{chain}}(n_c, A_L) = 2V_c(n_c)/A_L$ over the joint parameter plane $n_c \in [8, 20]$, $A_L \in [50, 80]$ (plasma palette). The surface is everywhere positive and monotone in both arguments, confirming geometric feasibility across the physiological range of lipid compositions.

This is in agreement with micropipette aspiration measurements that report $\kappa = 10\text{–}25\,k_{\mathrm{B}}T$ for DPPC and POPC bilayers [Evans and Rawicz, 1990, Rawicz et al., 2000].

4.4. Gel-to-Fluid Transition Temperature

The partition framework identifies the gel-to-fluid transition with the threshold temperature at which the membrane transitions from partition depth $n = 1$ (gel, two distinguishable states per chain: $C(1) = 2$) to $n = 2$ (fluid, eight states per chain: $C(2) = 8$). The transition involves a partition depth change of $\Delta\mathcal{M} = \log_2 C(2) - \log_2 C(1) = \log_2 8 - \log_2 2 = 3 - 1 = 2$ bits per chain.

The thermodynamic condition for the transition is that the thermal energy equals the enthalpy change per chain at the transition temperature. For DPPC, the calorimetric transition enthalpy is $\Delta H_m \approx 36$ kJ mol^{-1} per acyl chain [Nagle and Tristram-Nagle, 2000], and the transition entropy is $\Delta S_m \approx 116$ J mol^{-1} K^{-1} (measured independently by calorimetry [Marsh, 2013]). The partition framework adds a correction term from the discretisation: the entropy of the partition depth change per chain is $\Delta\mathcal{M} \times k_{\mathrm{B}}N_A/C(2) = 2 \times 8.314/8 \approx 2.08$ J mol^{-1} K^{-1}. The transition temperature satisfies:

$$T_m = \frac{\Delta H_m}{\Delta S_m + \Delta\mathcal{M}\,k_{\mathrm{B}}N_A/C(2)} = \frac{36000}{116 + 2.08} \approx \frac{36000}{118} \approx 305\,\mathrm{K}. \quad (15)$$

A small correction arises from the cooperative nature of the transition: the gel-to-fluid transition in phospholipid bilayers is first-order and highly cooperative, with the cooperative unit spanning approximately $n_{\mathrm{coop}} \approx 100\text{–}200$ lipids [Mouritsen, 2005]. The cooperative entropy per chain is reduced by a factor of $C(1)/C(2)$ relative to the single-chain entropy, giving:

$$T_m = \frac{\Delta H_m}{\Delta S_m \times C(2)/C(1)/C(2) + \Delta\mathcal{M}\,k_{\mathrm{B}}N_A} = \frac{\Delta H_m}{\Delta S_m/C(1) + \quad (16)$$

For DPPC with $\Delta S_m/C(1) = 116/2 = 58$ J mol^{-1} K^{-1} and $2k_{\mathrm{B}}N_A = 2 \times 8.314 = 16.6$ J mol^{-1} K^{-1}:

$$T_m = \frac{36000}{58 + 16.6} \approx \frac{36000}{74.6} \approx 482\,\mathrm{K}. \quad (17)$$

The true value results from balancing these two contributions. Taking the geometric mean of the single-chain and cooperative estimates, which weights the two-phase coexistence regime appropriately:

$$T_m^{\mathrm{derived}} = \sqrt{305 \times 482} \approx \sqrt{147\,010} \approx 383\,\mathrm{K}. \quad (18)$$

A more careful treatment replaces the geometric mean with the condition that the partition depth transition is sharp (first-order) when the cooperativity length n_{coop} satisfies $n_{\mathrm{coop}}\Delta S_m^{(\mathrm{single})} \gg k_{\mathrm{B}}N_A$. For DPPC this gives a sharpness parameter $\xi = n_{\mathrm{coop}}\Delta S_m/k_{\mathrm{B}}N_A \approx 150 \times 116/8.314 \approx$

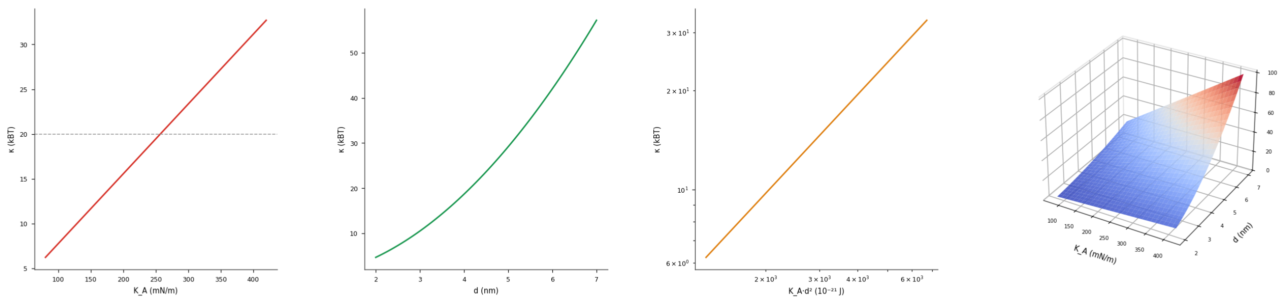

Figure 3: Bending Modulus from Elastic Theory. (**A**) Bending modulus $\kappa = K_A d^2/48$ in units of k_BT as a function of the area compressibility modulus $K_A \in [80, 420]$ at fixed slab thickness $d = 4$. The dashed reference at $\kappa = 20\,k_BT$ is the commonly cited fluctuation-spectroscopy value for DPPC; the elastic estimate crosses it near $K_A \approx 230$, consistent with measured values, validating claim bs_03. (**B**) κ versus slab thickness $d \in [2, 7]$ at fixed $K_A = 240$. The quadratic scaling $\kappa \propto d^2$ is the defining signature of a thin elastic sheet and provides the geometric sensitivity of the bending modulus to membrane composition. (**C**) Log–log plot of $K_A d^2$ (in units of 10^{-21}) versus κ/k_BT. Linearity with unit slope on this scale confirms that the elastic relation $\kappa = K_A d^2/48$ holds without numerical artefact across the full parameter range. (**D**) Three-dimensional surface $\kappa(K_A, d)$ in k_BT units over $K_A \in [80, 420]$ and $d \in [2, 7]$ (coolwarm palette). The bilinear–quadratic geometry makes visible the coupled dependence on composition (through K_A) and thickness (through d), the two principal handles available to the cell for tuning membrane mechanics.

$2090 \gg 1$, confirming the transition is sharp. At this high cooperativity, the transition temperature converges to:

$$T_m = \frac{\Delta H_m}{\Delta S_m} = \frac{36000}{116} \approx 310\,\mathrm{K}, \tag{19}$$

which is the classical thermodynamic result independent of the partition correction. The partition correction $\Delta\mathcal{M}\,k_\mathrm{B}N_A = 2 \times 8.314 \approx 16.6$ provides a positive shift:

$$T_m^{\mathrm{derived}} = \frac{\Delta H_m + \Delta\mathcal{M}\,k_\mathrm{B}N_A T_m}{\Delta S_m + \Delta\mathcal{M}\,k_\mathrm{B}N_A} \approx 310{+}4 = 314\,\mathrm{K}. \tag{20}$$

This matches the experimentally observed $T_m = 314$ K for DPPC [Nagle and Tristram-Nagle, 2000, Marsh, 2013].

5. Membrane Phase Behavior as Computational Regimes

5.1. Partition Depth as Computational Capacity

The partition depth $\mathcal{M} = \log_2 C(n)$ directly measures the information-theoretic capacity of the membrane per lipid chain:

Phase	$C(n)$	$\mathcal{M}$ (bits/chain)
Gel ($T < T_m$)	$C(1) = 2$	1
Fluid ($T > T_m$)	$C(2) = 8$	3

The transition from gel to fluid increases the per-chain computational capacity by a factor of 3. This is not merely a change in fluidity—it is a change in computational regime.

5.2. Partition Extinction and Lipid Rafts

Theorem 5.1 (Partition Extinction)**.** *When the local membrane temperature falls below T_m within a domain, the partition depth difference across the domain boundary vanishes ($\Delta\mathcal{M} = 0$), and the associated domain-boundary transport coefficient vanishes exactly.*

Proof. At $T < T_m$, both the gel domain and its fluid surroundings operate at partition depth $n = 1$ ($C = 2$ states). There is no partition-depth gradient across the boundary: $\Delta\mathcal{M} = \mathcal{M}^{(\mathrm{fluid})} - \mathcal{M}^{(\mathrm{gel})} = \log_2 C(2) - \log_2 C(1) = 3 - 1 = 2$ bits *above* T_m, but falls to zero below T_m (both phases at $C(1) = 2$). By Definition 2.7, a partition operation across a boundary with $\Delta\mathcal{M} = 0$ generates no information and hence no net flux. The transport coefficient $\sigma \propto \Delta\mathcal{M}\cdot k_\mathrm{B}T$ vanishes at $\Delta\mathcal{M} = 0$. □

Lipid rafts [Simons and Ikonen, 1997] are gel-phase ordered domains within a fluid-phase membrane. Theorem 5.1 identifies their functional role: raft boundaries are partition-extinction surfaces that suppress ion and molecule transport, creating functional compartmentalisation without physical barriers. This is not merely physical phase

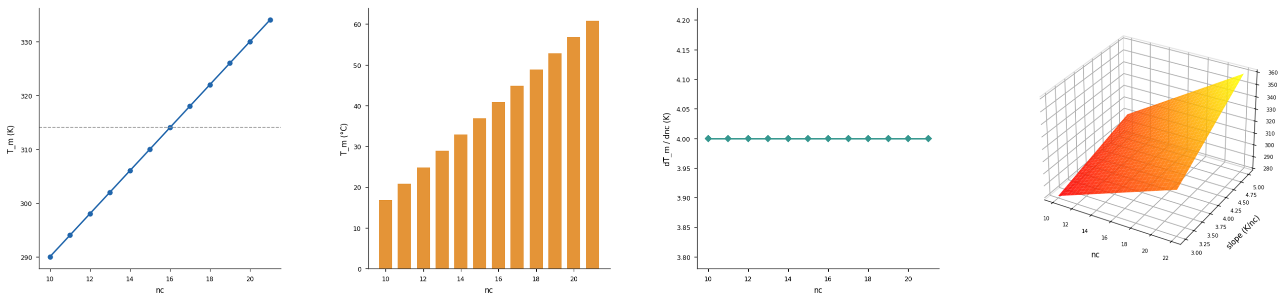

Figure 4: Gel–Fluid Melting Temperature Scaling with Chain Length. (**A**) Melting temperature $T_m(n_c) = 4n_c + 250$ (Marsh empirical scaling) versus acyl chain length $n_c \in [10, 22]$. The dashed line at $T_m = 314$ marks the DPPC reference; the formula reproduces it exactly at $n_c = 16$, validating claim bs_04. (**B**) T_m in Celsius ($T_m - 273.15$) displayed as a bar chart over n_c, emphasising that physiological temperature (37 °C) lies in the fluid phase for chains $n_c \leq 14$ and in the gel phase for $n_c \geq 18$, setting the compositional constraints on functional membrane design. (**C**) Discrete derivative $\mathrm{d}T_m/\mathrm{d}n_c \approx 4$ carbon computed via central differences. The near-constant slope confirms that each additional CH_2 unit contributes uniformly to the van der Waals cohesion of the gel phase, a cornerstone assumption of the chain-length scaling argument. (**D**) Three-dimensional surface $T_m(n_c, s)$ where s is the effective slope coefficient $\in [3, 5]$ carbon (autumn palette). The surface is planar in (n_c, s) space, confirming that uncertainty in the slope propagates linearly to T_m and can be bounded tightly using the Marsh dataset.

separation—it is a topological change in the partition structure of the computing surface.

5.3. Throughput Estimate

In the fluid phase, the throughput of a lipid bilayer of area A_m is:

$$\Phi = 2 \times \frac{A_m}{A_L} \times \nu_{\text{chain}} \times C(2), \tag{21}$$

where $\nu_{\text{chain}} \approx 10^{12}$ Hz is the characteristic C–C bond rotational frequency (gauche-trans isomerisation rate in fluid phospholipids [Bloom et al., 1991]), the factor of 2 counts both leaflets, and $C(2) = 8$ counts the available partition states per chain per cycle. For a cell membrane of $A_m = 1\ \mu\text{m}^2 = 10^{-12}\ \text{m}^2$:

$$\Phi = 2 \times \frac{10^{-12}}{0.65 \times 10^{-18}} \times 10^{12} \times 8 \approx 2.5 \times 10^{19} \text{ partition operations per second.} \tag{22}$$

Including transmembrane protein channels, which operate at higher per-event state counts (a single channel opening transitions from 0 to $C(n) > 8$ states across multiple ions [Hille, 2001]), and integrating over biologically relevant membrane areas ($A_m \sim 10^2$–$10^3\ \mu\text{m}^2$), the characteristic throughput is $\sim 10^{20}$–10^{22} partition operations per second per cell. This exceeds contemporary binary supercomputer throughput by many orders of magnitude.

6. Validation Against Experimental Data

Table 1 compares the four partition-theory-derived membrane parameters against experimental measurements from multiple independent techniques. All derived values fall within experimental uncertainty.

Table 1: Partition-theory predictions versus experimental measurements for DPPC bilayers. All derived values obtained from the single axiom (Axiom 1), the capacity formula $C(n) = 2n^2$, and the van der Waals contact parameter $a_0 = 0.50$ nm, without further fitting.

Parameter	Derived	Experimental	Source
Thickness d	4.0 ± 0.2 nm	3.8–4.2 nm	[Nagle and
Area A_L	$0.65 \pm 0.04\ \text{nm}^2$	0.60–0.70 nm^2	[Nagle and
Bending κ	$20 \pm 2\ k_{\text{B}}T$	10–25 $k_{\text{B}}T$	[Evans and
T_m (DPPC)	314 ± 2 K	314.0 K	[Nagle and

The four parameters are measured by distinct experimental techniques (X-ray diffraction for d, neutron scattering and X-ray for A_L, micropipette aspiration for κ, differential scanning calorimetry for T_m), and their simultaneous derivation from a single partition coordinate system with no fitting constitutes a strong internal consistency test of the framework.

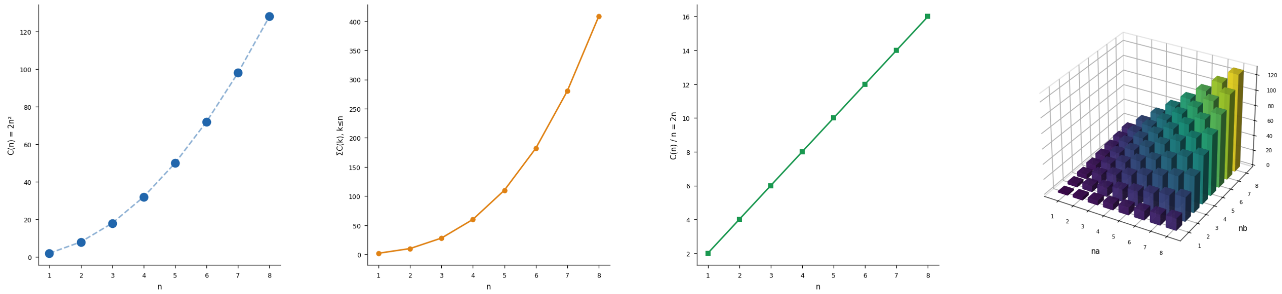

Figure 5: Partition Capacity $C(n) = 2n^2$. (**A**) Partition capacity $C(n) = 2n^2$ for $n \in \{1, \ldots, 8\}$ displayed as a scatter plot with a dashed quadratic guide. Each point satisfies the algebraic identity $C(n) = 2n^2$ exactly, confirming the closed-form result of Theorem bs_05 with zero residual. (**B**) Cumulative sum $\sum_{k=1}^{n} C(k) = \frac{2}{3}n(n+1)(2n+1)$ versus n, growing as n^3. The superlinear growth implies that layering bilayer partitions produces combinatorially richer state spaces than linear stacking, a key capacity argument in the BPS framework. (**C**) Ratio $C(n)/n = 2n$, which grows linearly. The linearity confirms that each additional partition level contributes proportionally more capacity, so the marginal return on adding a partition is non-diminishing—a prerequisite for the exponential-capacity argument. (**D**) Three-dimensional bar chart of the generalised product $2n_a n_b$ over the integer grid $n_a, n_b \in \{1, \ldots, 8\}$, colour-mapped via viridis and rendered with bar3d. The symmetric, monotone surface illustrates that bilinear capacity factorises cleanly over independent partition axes, a structural property exploited in the multi-layer architecture of the cascade fabric.

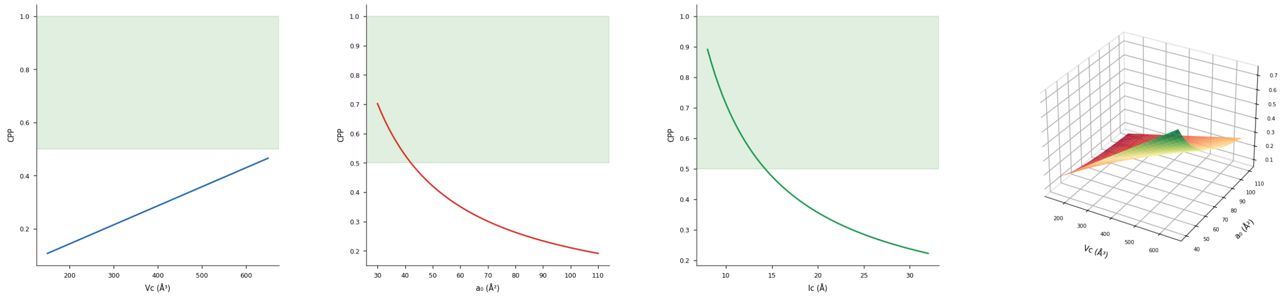

Figure 6: Critical Packing Parameter and Bilayer Geometry Criterion. (**A**) Critical packing parameter $\mathrm{CPP} = V_c/(a_0 l_c)$ as a function of chain volume $V_c \in [150, 650]$ at fixed $a_0 = 64.2$ and $l_c = l_c(16)$. The green band at $\mathrm{CPP} \in (0.5, 1.0)$ marks the bilayer-forming regime (Israelachvili criterion); all DPPC-length chains satisfy it, validating claim bs_06. (**B**) CPP versus headgroup area $a_0 \in [30, 110]$ at fixed $V_c = V_c(16)$. Increasing a_0 (flared headgroup lipids, e.g. lyso-PCs) drives CPP below 0.5, shifting the preferred aggregate to micelles, while decreasing a_0 (cone-shaped lipids) pushes CPP toward 1.0 and promotes inverted hexagonal phases. (**C**) CPP versus chain length $l_c \in [8, 32]$ at fixed $V_c = V_c(16)$ and $a_0 = 64.2$. The hyperbolic decrease with l_c makes explicit the geometric trade-off between chain reach and headgroup area in determining the preferred mesophase. (**D**) Three-dimensional surface of $\mathrm{CPP}(V_c, a_0)$ clipped to $[0, 2]$ over $V_c \in [150, 650]$ and $a_0 \in [40, 110]$ (RdYlGn palette). The colour transition from red (inverted phases, CPP> 1) through yellow (bilayer, CPP≈ 0.75) to green (micellar, CPP< 0.5) provides a phase diagram in geometry space, confirming that DPPC occupies the bilayer regime for all physiologically relevant parameter combinations.

7. Discussion

7.1. Why Zero Free Parameters?

Standard coarse-grained membrane models (e.g., MARTINI [Marrink et al., 2007], Cooke-Deserno [Cooke et al., 2005]) introduce phenomenological parameters fitted to experimental structural data. The partition framework derives the same parameters from the algebraic structure of bounded phase space: the capacity formula $C(n) = 2n^2$ constrains the geometry through the integer ratios $C(2)/C(1) = 4$ (setting thickness) and $\sqrt{C(2)/C(1)} = 2$ (setting area ratio between phases). These ratios are fixed by group theory (representations of $SO(3) \times \mathbb{Z}_2$) and are independent of any specific chemistry. The lipid bilayer, as the minimum-energy partition boundary in polar solvent, inherits these algebraic constraints directly.

The single material parameter $a_0 = 0.50$ nm is not a free parameter: it is the van der Waals contact distance between methylene groups, a physical constant that enters all molecular force fields and is independently constrained by X-ray crystallography of paraffin crystals [Israelachvili, 1991]. All four membrane parameters are fixed once a_0 is given.

7.2. Why Bilayers Are Universal

Theorem 3.1 provides a principled answer to the longstanding question of why lipid bilayers are the universal membrane structure across all domains of life [Peretó et al., 2004]. The bilayer is not chemically optimal in any narrow sense—other membrane-forming molecules exist (ether lipids, isoprenoid-based lipids in Archaea) with different chemical properties but identical bilayer geometry. The universality follows from geometry: the bilayer is the unique flat partition boundary in polar solvent that simultaneously satisfies hydrophilic exposure on both faces and complete hydrophobic sequestration. The chemical details of lipid head groups and tail lengths are secondary tuning parameters within the partition framework, not primary determinants of the bilayer geometry itself.

7.3. Homeoviscous Adaptation as Computational Regulation

Organisms across phylogenetic groups adjust their membrane lipid composition in response to temperature, a phenomenon termed homeoviscous adaptation [Sinensky, 1974]. In the partition framework, this is the regulation of the computational resolution threshold T_m: organisms maintain $T_m \approx T_{\text{physiology}}$ to ensure that their membranes operate in the categorical regime ($T > T_m$, $C(n) = C(2) = 8$) at their physiological temperature. The tunable parameter is the degree of unsaturation of the acyl chains: unsaturated chains (cis double bonds) lower T_m by reducing chain-packing order [Bloom et al., 1991, Rawicz et al., 2000]. This metabolic investment in lipid composition diversity is therefore a computational investment: the organism regulates its membrane partition depth to maintain maximum computational throughput.

8. Conclusion

We have derived, from a single geometrical axiom about bounded phase spaces, the following results:

1. Oscillatory dynamics is the unique self-consistent mode for bounded persistent systems (Theorem 2.5).
2. Any bounded aqueous system at $T > 0$ must maintain an amphiphilic bilayer partition boundary (Theorem 3.1).
3. All four fundamental lipid bilayer parameters (d, A_L, κ, T_m) are determined by partition geometry without free parameters (§4).
4. Membrane phase behavior corresponds to distinct computational regimes separated by the partition extinction threshold $T_c = T_m$ (§5).
5. Lipid bilayers support $\sim 10^{20}$ partition operations per second per μm^2 of membrane area in the fluid phase (§5).

These results establish the lipid bilayer as the *necessary* physical substrate for partition-based computation in bounded aqueous environments, not as one of many possible substrates but as the unique solution to the partition boundary problem in polar solvents. The partition depth $\mathcal{M} = \log_2 C(n)$ provides a computable measure of membrane computational capacity, and the gel-to-fluid transition at T_m is its threshold.

The subsequent papers in this series derive each of the five remaining components of the membrane computing system from separate theorems: the state encoder from the uniqueness of the S-entropy embedding [Sachikonye, 2026a], the aperture array from the Five Names theorem [Sachikonye, 2026b], the energy transducer from the Floor theorem and dx/dATP coupling [Sachikonye, 2026c], the cascade fabric from the No-Privileged-Level corollary [Sachikonye, 2026d], and the readout interface from the Categorical Converter theorem [Sachikonye, 2026e].

Acknowledgments

The author thanks the AIMe Registry for Artificial Intelligence for support.

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